In [2]:
```
import pandas as pd
import numpy as np
# Load the dataset
df =
pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/datasets/mast
er/titanic.csv')
# Let's look at the missing values
print("--- Missing Values Before Cleaning ---")
print(df.isnull().sum())
# --- ENTER YOUR CODE HERE ---
# 1. Calculate the median of the 'Age' column
median_age = df['Age'].median()

# 2. Fill the missing values in 'Age' with the median
df['Age'] = df['Age'].fillna(median_age)

# 3. Verify that there are no more missing values in 'Age'
print("Missing values in 'Age' after imputation:")
print(df['Age'].isnull().sum())

# 1. Use get_dummies to create new columns for 'Sex' and 'Embarked'
encoded_df = pd.get_dummies(df, columns=['Sex', 'Embarked'],
drop_first=True)

# 2. Display the first few rows of the new DataFrame to see the new columns
print(encoded_df.head())

from sklearn.preprocessing import StandardScaler

# 1. Create an instance of the StandardScaler
scaler = StandardScaler()

# 2. Select the columns to scale
columns_to_scale = ['Age', 'Fare']

# 3. Fit the scaler to the data and transform it
encoded_df[columns_to_scale] =
scaler.fit_transform(encoded_df[columns_to_scale])

# 4. Display the first few rows to see the scaled data
print(encoded_df.head())
```

```
--- Missing Values Before Cleaning ---
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
```

```
Fare             0
Cabin          687
Embarked         2
dtype: int64
Missing values in 'Age' after imputation:
0
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                                 Name   Age  SibSp  Parch  \
0                             Braund, Mr. Owen Harris  22.0      1      0
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  38.0      1      0
2                              Heikkinen, Miss. Laina  26.0      0      0
3        Futrelle, Mrs. Jacques Heath (Lily May Peel)  35.0      1      0
4                            Allen, Mr. William Henry  35.0      0      0

             Ticket     Fare Cabin  Sex_male  Embarked_Q  Embarked_S
0         A/5 21171   7.2500   NaN      True       False        True
1          PC 17599  71.2833   C85     False       False       False
2  STON/O2. 3101282   7.9250   NaN     False       False        True
3            113803  53.1000  C123     False       False        True
4            373450   8.0500   NaN      True       False        True
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                                 Name       Age  SibSp  Parch
\
0                             Braund, Mr. Owen Harris -0.565736      1      0
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  0.663861      1      0
2                              Heikkinen, Miss. Laina -0.258337      0      0
3        Futrelle, Mrs. Jacques Heath (Lily May Peel)  0.433312      1      0
4                            Allen, Mr. William Henry  0.433312      0      0

             Ticket      Fare Cabin  Sex_male  Embarked_Q  Embarked_S
0         A/5 21171 -0.502445   NaN      True       False        True
1          PC 17599  0.786845   C85     False       False       False
2  STON/O2. 3101282 -0.488854   NaN     False       False        True
3            113803  0.420730  C123     False       False        True
4            373450 -0.486337   NaN      True       False        True
```

Knowledge Check Answers

1. Why is it often better to impute missing values with the median instead of the mean?

The median is far less sensitive to extreme outliers or skewed distributions than the mean. The mean gets pulled in the direction of unusually high or low values, which can introduce bias into the dataset. The median represents the middle value, making it a much safer and more robust estimate for skewed numerical data.

2. Explain in your own words what One-Hot Encoding does and why it is necessary.

Machine learning algorithms rely on mathematical formulas and operate on numerical values, so they cannot directly interpret non-numeric text strings like 'male' or 'female'. One-Hot Encoding converts categorical text values into distinct numerical columns containing binary indicators (1 or 0). This allows the model to process categorical information correctly without accidentally implying an artificial numerical ranking between categories.

3. Would you need to apply Feature Scaling to a Decision Tree model? Why or why not?

No, feature scaling is unnecessary for Decision Tree models. Decision Trees make predictions by evaluating one feature at a time through threshold-based splits (e.g., Age > 30 or Fare > 50). Because each split decision depends only on the relative ordering of values within an individual column rather than a combined distance calculation across features, scaling the numerical ranges has no impact on where or how the tree splits.